XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE407 Electrical Energy Utilization

COURSE OVERVIEW

By:
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Phone: -
Consultation Hour: by appointment

Programme Educational Objective (PEO)

PEO1	**Professionalism** Graduates will be capable of using problem-solving techniques and critical analysis to establish themselves as practicing professionals and demonstrate leadership skills and entrepreneurial spirit in the field of Electrical and Electronics Engineering or related fields.
PEO2	**Continuous Personal Development** Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.
PEO3	**Societal Engagement** Graduates will contribute to sustainable development and the well-being of society.

Course Learning Outcome (CLO) and Programme Learning Outcome (PLO)

CLO1	Explain the important aspects of efficient electrical energy utilization.
CLO2	Analyze the problems related to the economic aspects of demand side management, efficient energy equipment, and renewable energy generation.
CLO3	Evaluate energy saving solution based on given energy data.

PLO10	Communication - Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PLO2	Problem Analysis - Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences

Course Assessment

Assessment	Detail	(%)
Continuous Assessment	Assignment (Group) - Report - Presentation	30
	Quiz (Individual)	20
Final Exam (2 hours) Answer All Questions	4 Questions	50
	Total	100

Grading

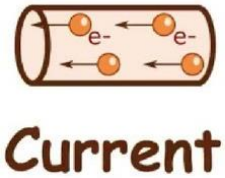
Exam Questions, Marking Scheme & Invigilation	
Special Requirements (For EAC approved Engineering Programmes)	• Multiple Choice Questions (MCQ) are not encouraged in the final exams. • **Students shall pass the final examination to pass the course, the threshold marks for final examination is 40.**

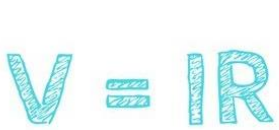
Marks	Grade	Grade Points	Description
85 – 100	A	4.00	Distinction
80 – 84	A-	3.70	Distinction
75 – 79	B+	3.30	Good
70 – 74	B	3.00	Good
65 – 69	B-	2.70	Good
60 – 64	C+	2.30	Pass
55 – 59	C	2.00	Pass
50 - 54	C-	1.70	Conditional Pass
45 – 49	D+	1.30	Fail
40 - 44	D	1.00	Fail
≤ 39	F	0.00	Fail

Grading scales and grades applicable to ALL intakes effective from Academic session 2017/09

Course outline

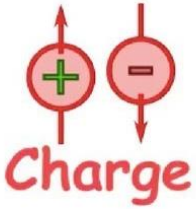
1. Introduction to energy management and conservation
    I. Energy basics
    II. Sustainability principles in management
    III. Electricity consumption in buildings
    IV. Energy management program











Course outline

2. Energy audit for building
   I. General procedure for a detailed energy audit
   II. Measurement and verification of energy savings
   III. Government initiatives



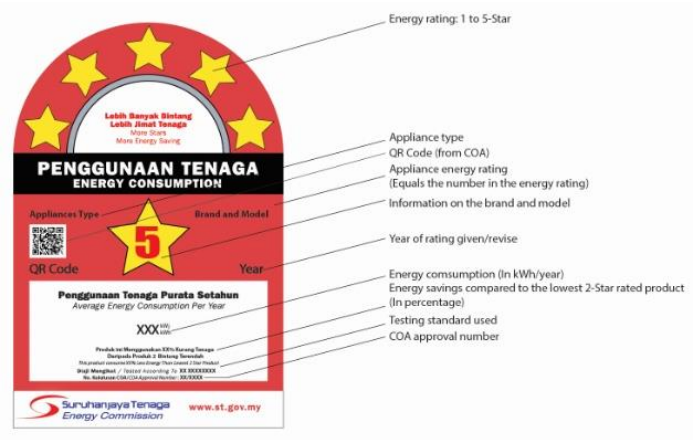
Course outline

3. Energy management
   I. Policies, standards & regulations
   II. Demand side management
   III. Electrical hazards and safety
   IV. Wiring circuit and electrical installation system



Course outline

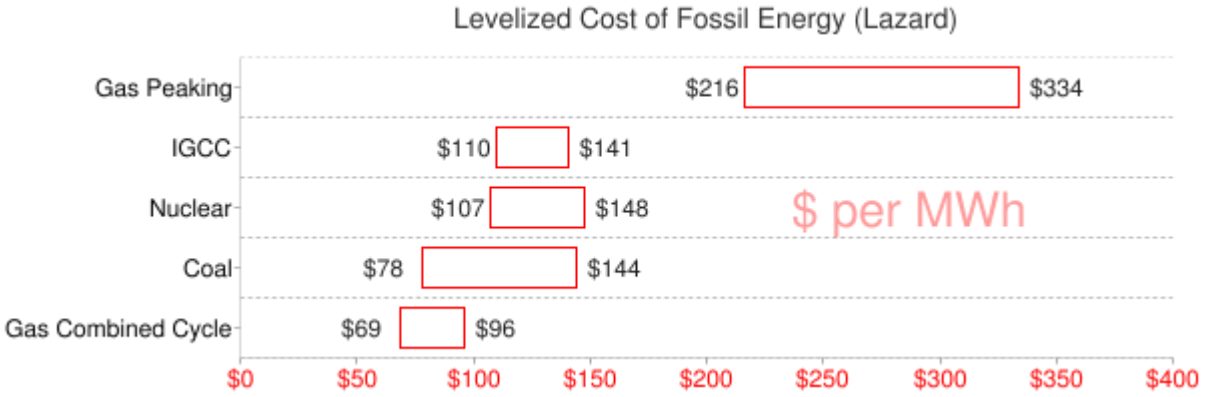
4. Energy efficient equipment
   - I. Air-conditioning & mechanical ventilation system
   - II. Electric heating, arc furnaces and electric welding
   - III. Lighting system
   - IV. Other equipment & appliances





Course outline

5. Economics of distributed resources
   I. Energy economics
   II. Levelized busbar costs

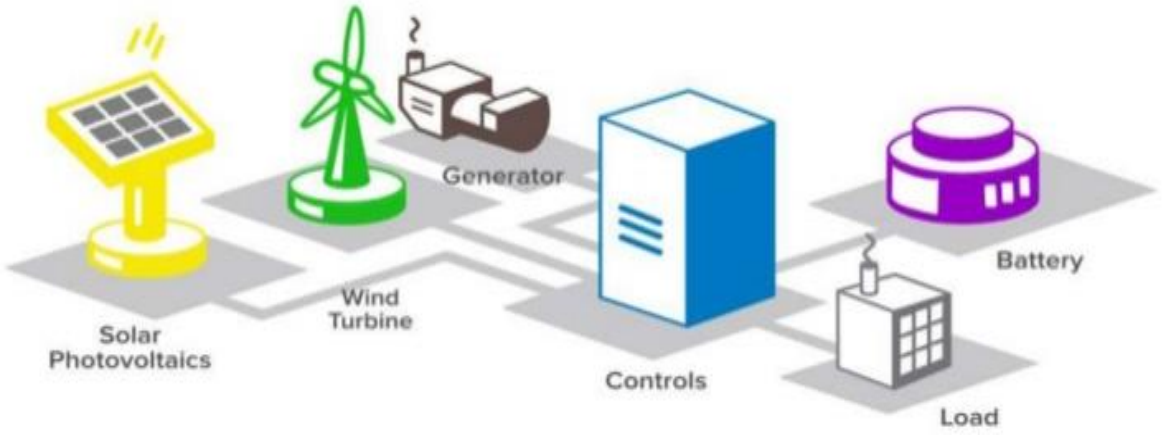


Course outline

6. Distributed generation

DG technologies

I. Fuel cell
II. Biomass
III. Micro-hydropower
IV. Photovoltaic
V. Wind turbine



Attendance

- Attendance is required by the Ministry of Higher Education Malaysia.
- Sign the attendance sheet during class.
- If you cannot make it to class, sick leaves and other types of leaves should be applied 3 days prior/after, else you will be marked as absent.
- For two consecutive absences, you will be reported to Academic Affairs.

Course communication

- Email or Teams message
  - Requests for extensions, questions about course policies
  - Grading inquiries, requests for help with assignments
  - Sick leaves and other types of leaves
- Face-to-face
  - Class breaks
  - Email/message me to make an appointment for consultations

References

- Required reference
  - Frank Kreith and D. Yogi Goswami, Energy Management and Conservation Handbook, CRC Press, 2017.
- Further readings
  - Wayne C. Turner, Energy Management Handbook, Fairmont Press Inc, 2017.
  - Kementerian Tenaga, Teknologi Hijau dan Air, The Energy Efficiency and Conservation Guidelines Part 1: Electrical Energy - use Equipment, 2011
  - Brian Scaddan, Electrical Installation Work, 9th Edition, Routledge, 2018